Knowledge Representation & Intelligent Agents

Project 3.1: Intelligent Campus Navigation Agent

Problem Statement:
Design an intelligent agent that assists students in navigating a university campus by reasoning over spatial knowledge, schedules, and preferences.

Key Concepts:

- Knowledge Representation
- Intelligent Agents
- Multi-agent Communication

Technologies & Frameworks:

- Python
- Knowledge Graphs
- Agent-based modeling using **JADE**
- JSON / RDF

Outcome:
An autonomous agent capable of reasoning, negotiating routes, and providing optimal navigation guidance.

3.2: Trust-Based Multi-Agent E-Commerce System

Problem Statement:
Develop a multi-agent e-commerce platform where buyer agents evaluate seller agents based on trust, reputation, and negotiation strategies.

Key Concepts:

- Trust & Reputation Models
- Agent Negotiation & Bargaining

Technologies & Frameworks:

- Python
- Flask
- Multi-agent simulation
- Rule-based reasoning

Outcome:
A simulation demonstrating intelligent negotiation and trust-aware decision-making.

Logic Programming & Slot–Filler Structures

Project 4.1: AI-Based Legal Reasoning System

Problem Statement:
Build a logic-based system that determines legal eligibility (e.g., bail or loan approval) using propositional and first-order logic.

Key Concepts:

- Propositional Logic
- First-Order Logic
- Forward & Backward Chaining

Technologies & Frameworks:

- **SWI-Prolog**
- Python (for interface)

Outcome:
An explainable AI system capable of logical inference and reasoning.

Project 4.2: Medical Diagnosis Using Semantic Networks

Problem Statement:
Create a medical knowledge base using semantic networks and frames to assist in preliminary disease diagnosis.

Key Concepts:

- Semantic Networks
- Frames & Slot–Filler Structures

Technologies & Frameworks:

- Python
- NetworkX
- Neo4j Graph Database

Outcome:
A structured and interpretable diagnostic support system.

Planning, Learning & Game Playing

Project 5.1: AI-Based Automated Travel Planner

Problem Statement:
Design an AI planner that generates optimal travel plans based on user goals, constraints, and available actions.

Key Concepts:

- STRIPS
- Goal Stack Planning
- Knowledge-Based Planning

Technologies & Frameworks:

- Python
- PDDL
- Planning Domain Simulators

Outcome:
An automated planner that generates feasible and optimized plans.

Project 5.2: Intelligent Game Playing Agent

Problem Statement:
Develop an AI agent that plays a strategy game using Minimax with Alpha-Beta Pruning.

Key Concepts:

- Game Trees
- Minimax Algorithm
- Alpha-Beta Pruning

Technologies & Frameworks:

- Python
- Pygame

Outcome:
An optimal game-playing agent with performance evaluation.

Advanced AI Topics

Project 6.1: AI-Based Crop Disease Prediction System

Problem Statement:
Build a neural-network-based system to classify crop diseases from leaf images.

Key Concepts:

- Neural Networks
- Supervised Learning

Technologies & Frameworks:

- **TensorFlow**
- **Keras**
- Python

Outcome:
An AI-powered decision support tool for agriculture.

Project 6.2: Fuzzy Logic-Based Smart Traffic Signal System

Problem Statement:
Design a fuzzy inference system to dynamically control traffic signals based on vehicle density.

Key Concepts:

- Fuzzy Sets
- Fuzzy Rules
- Decision Systems

Technologies & Frameworks:

- Python
- **scikit-fuzzy**

Outcome:
An adaptive traffic control system improving traffic flow efficiency.

Genetic Algorithm for Timetable Optimization

Problem Statement:
Apply genetic algorithms to generate conflict-free university timetables with optimal resource utilization.

Key Concepts:

- Genetic Algorithms
- Optimization
- Fitness Functions

Technologies & Frameworks:

- Python
- **DEAP**

Outcome:
An optimized scheduling system using evolutionary computation.